Avionics for Model Rockets

AEROSPACE

ENGINEERING CLUB

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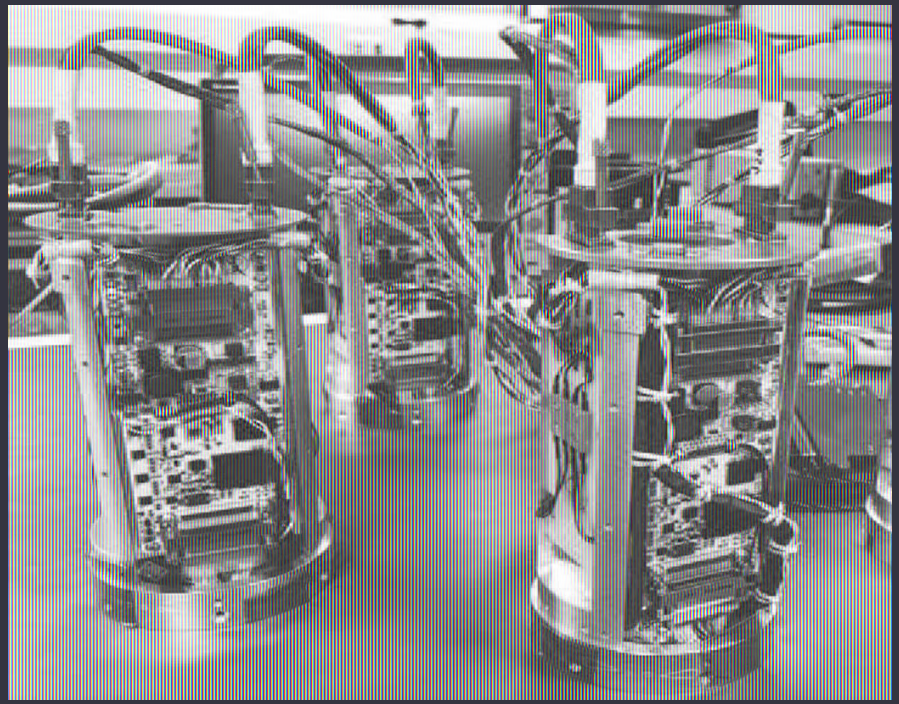
Introduction

As someone who had made a nice model rocket, wouldn’t you like to(...)

- Analyze your flight data after a successful/terrible rocket launch?
- Ensure flight safety and mission success?
- Have State Awareness?
- Upgrade your rocket by introducing a method to stabilize it during flight?

Introducing... Avionics!



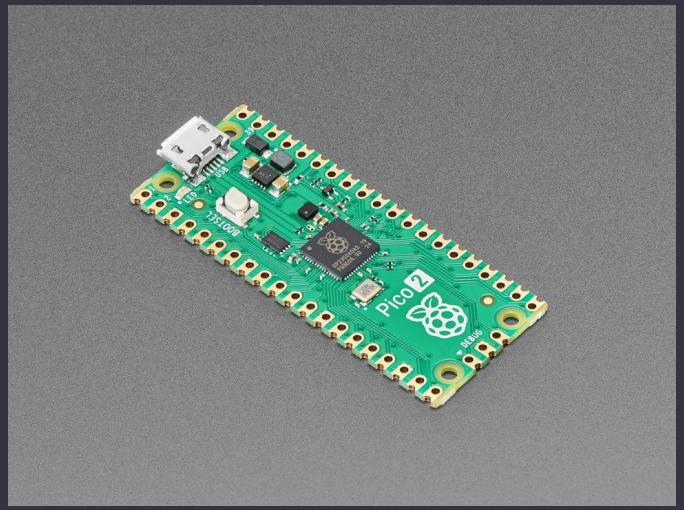
An Avionics Bay requires these key things:

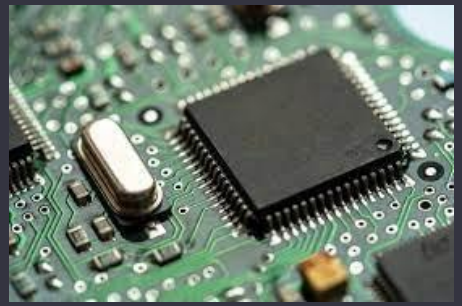
1. Power System

- It all starts with a battery. There are a few decisions, based on the layout of your Avionics bay, that must be made:
  - What Type?(Li-Po, Li-ion, Alkaline)
  - Voltage Regulation(3.3V Versus 5V ground rails)
  - Power Distribution
  - Note: Power Stability matters, especially at high G's

2. Flight Computer/Controller

- The Brain
- Executes flight logic(Guidance, Stability,etc)
- Reads sensors, logs data, fires outputs

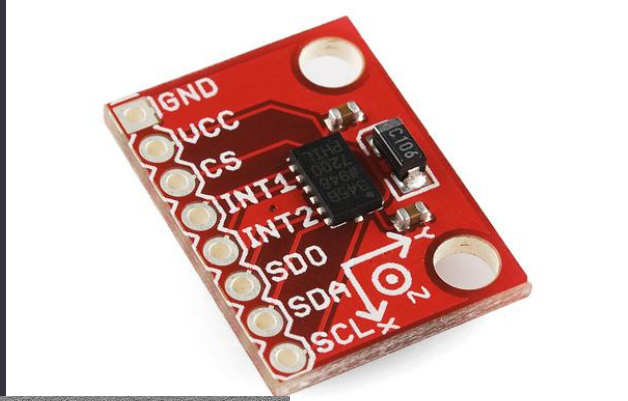


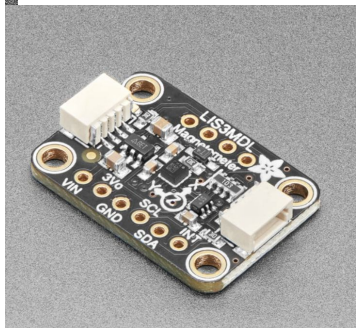


3. Sensors

- These tell the rocket what it’s doing

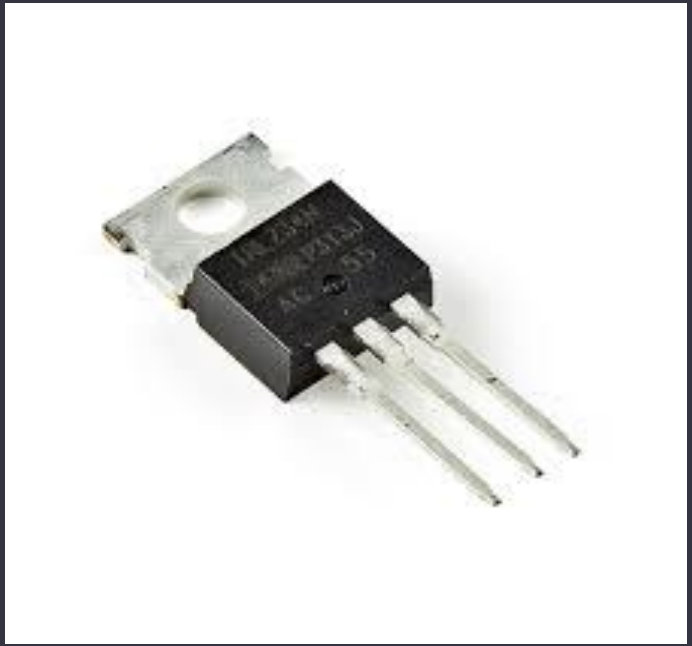
- Common Sensors:
  - Accelerometer
  - Gyroscope
  - Barometric Pressure Sensor
  - GPS
  - Magnetometer





4. Interesting thingy

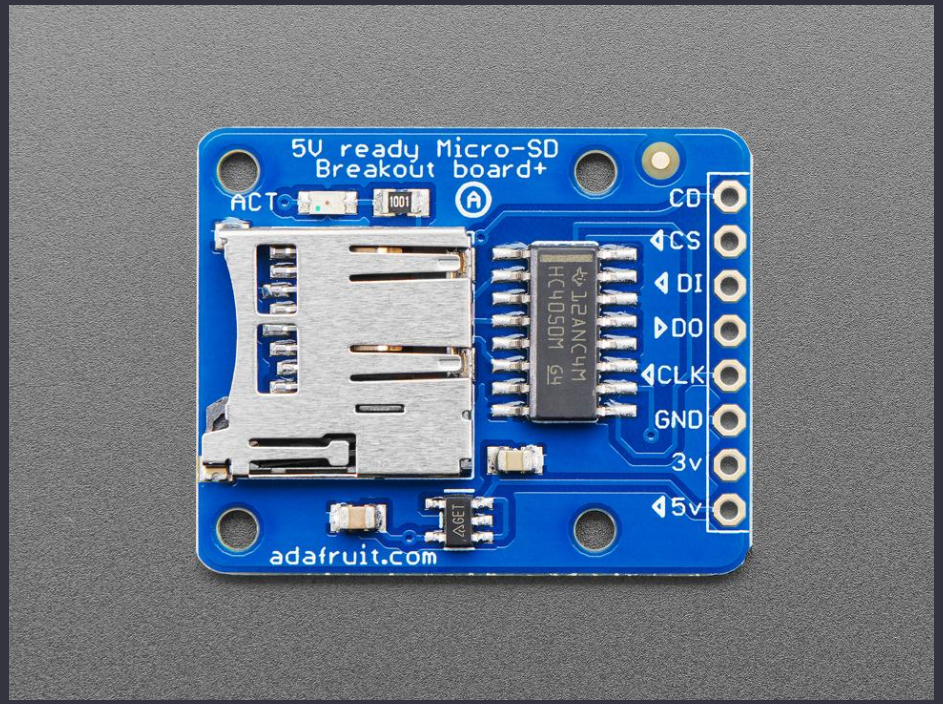
- MOSFETS(Current-controlling switches)



5. Data-Handling

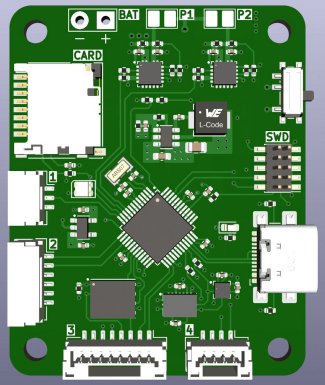
- Onboard Storage(SD CARDS)

- Optional: Telemetry Radio

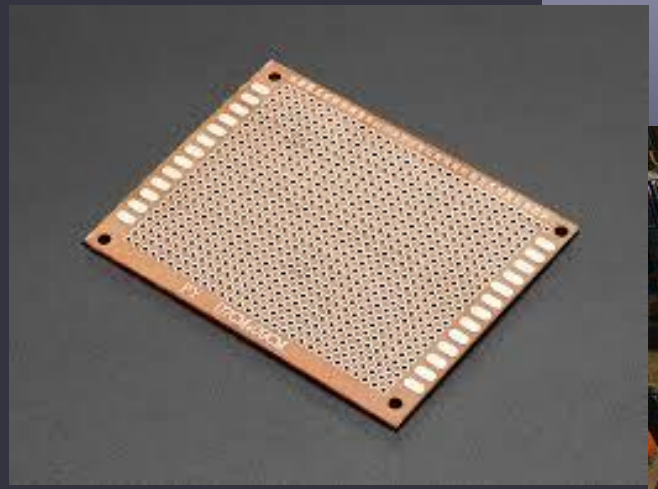


6. Your Base

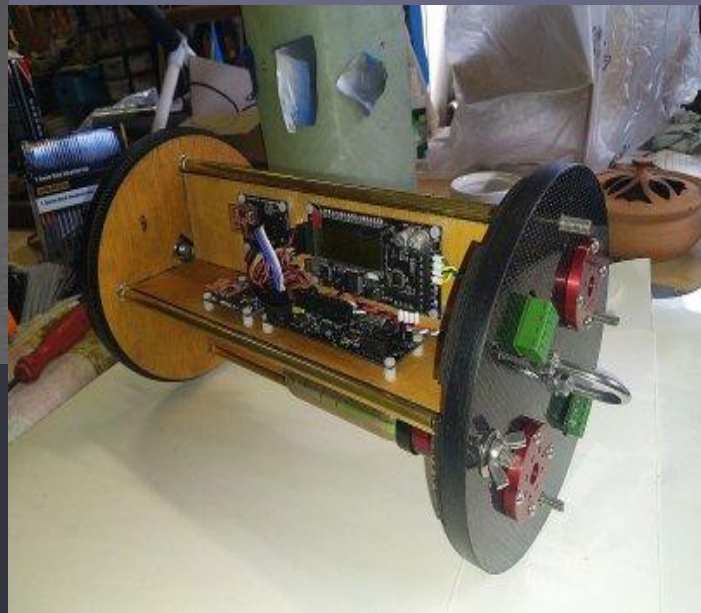
- PCB



- Or Perfboard

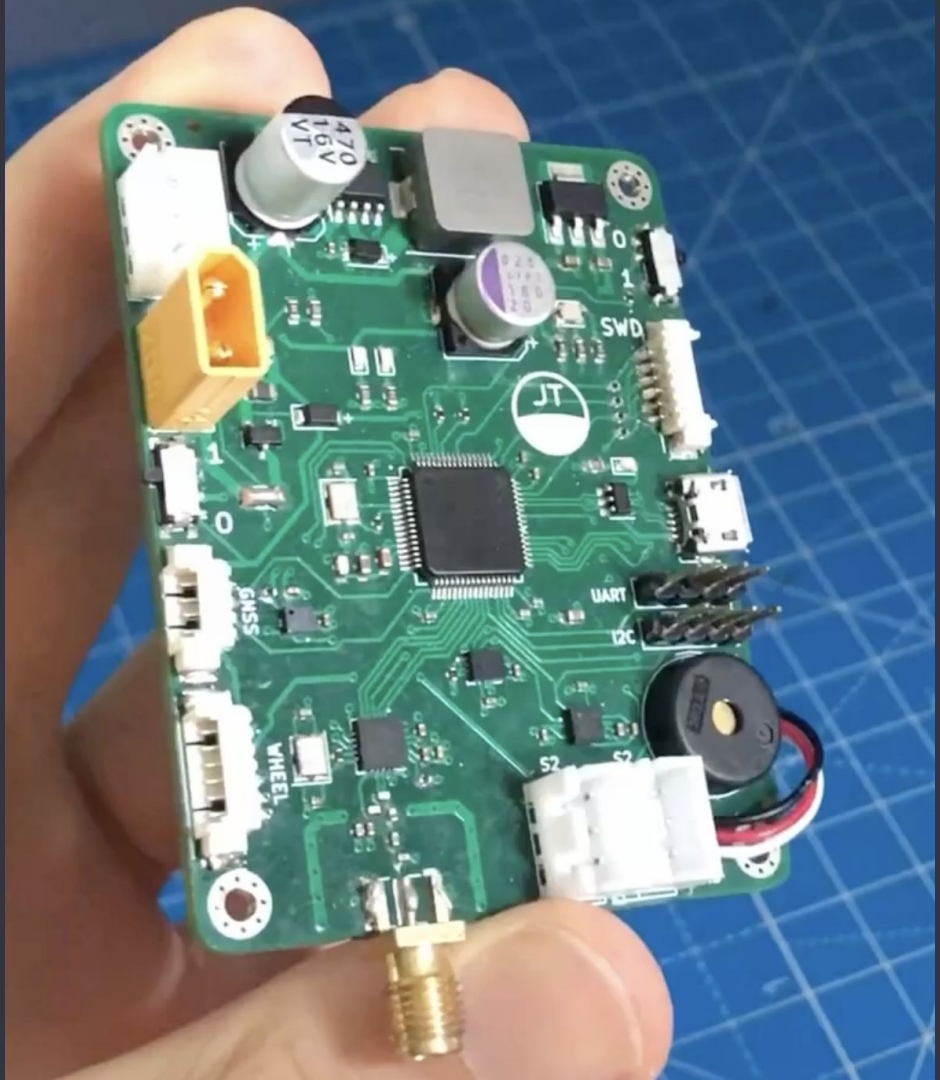


- Or Neither?



Examples

OMEGA AVIONICS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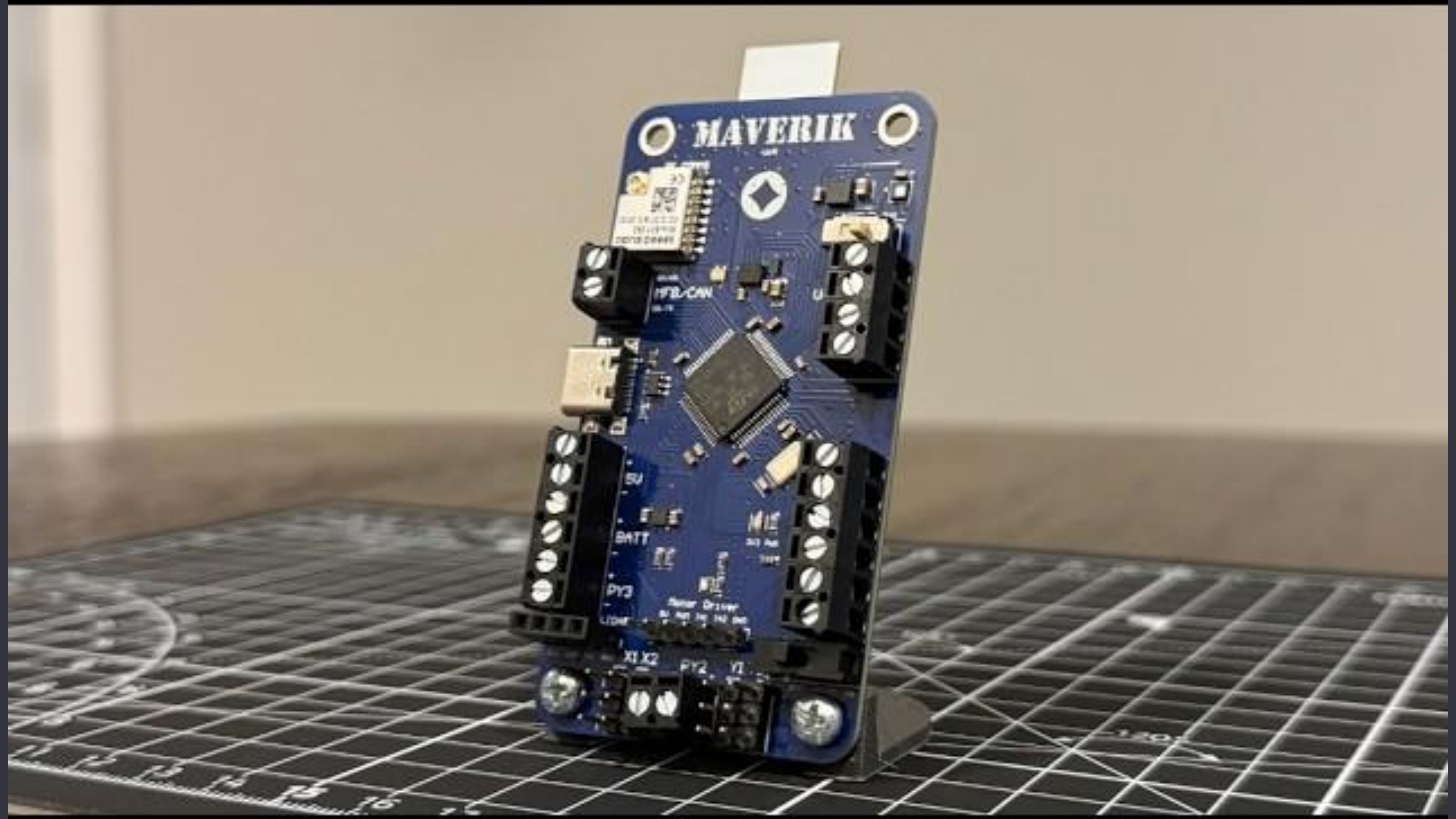
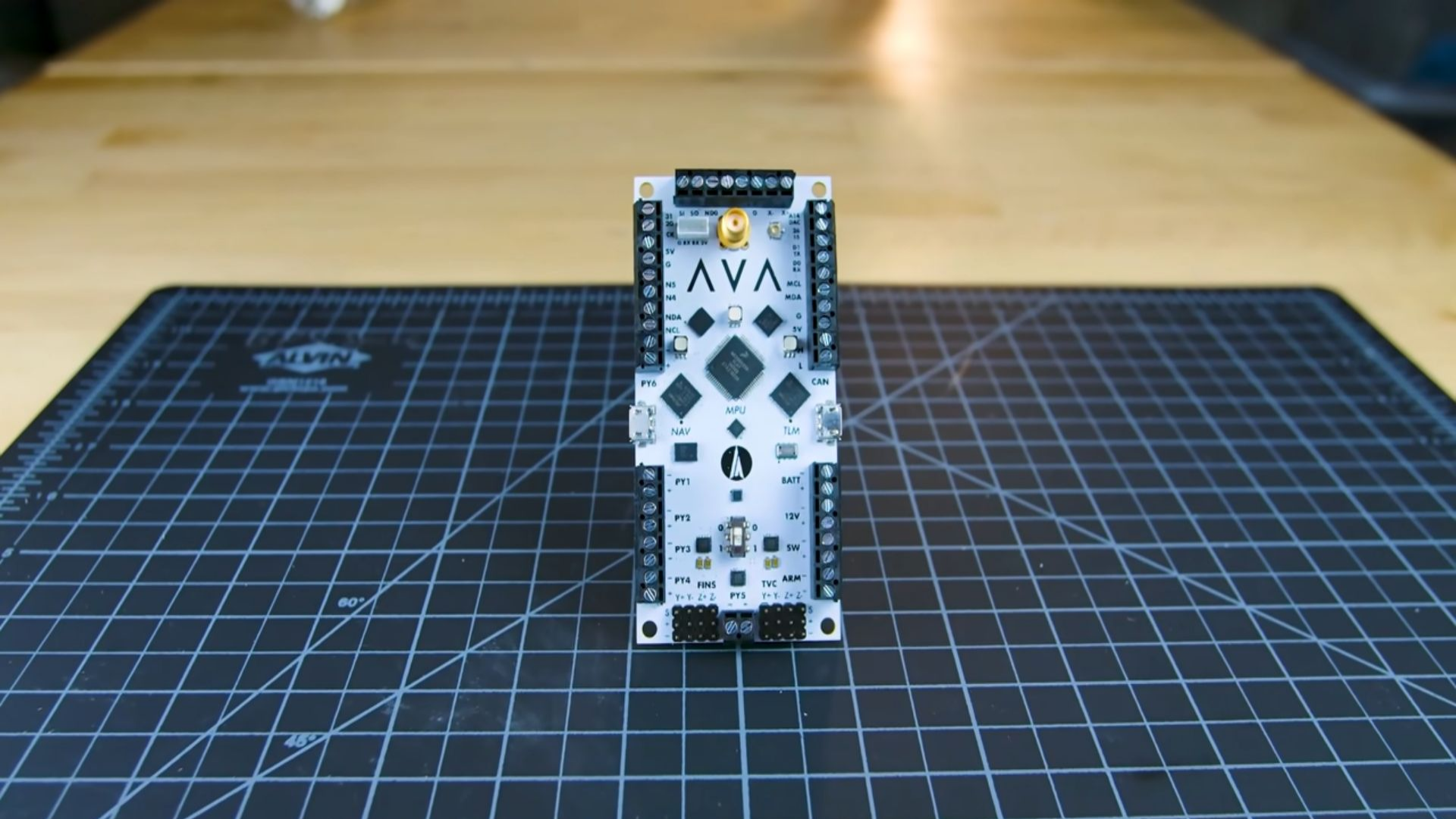
JT
GNSS
WHEEL
UART
I2C
S2
S2


MAVERIK
HFB/CAN
BATT
PY3
X1 X2 PY2 Y1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PY3
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12V
SW
ARM
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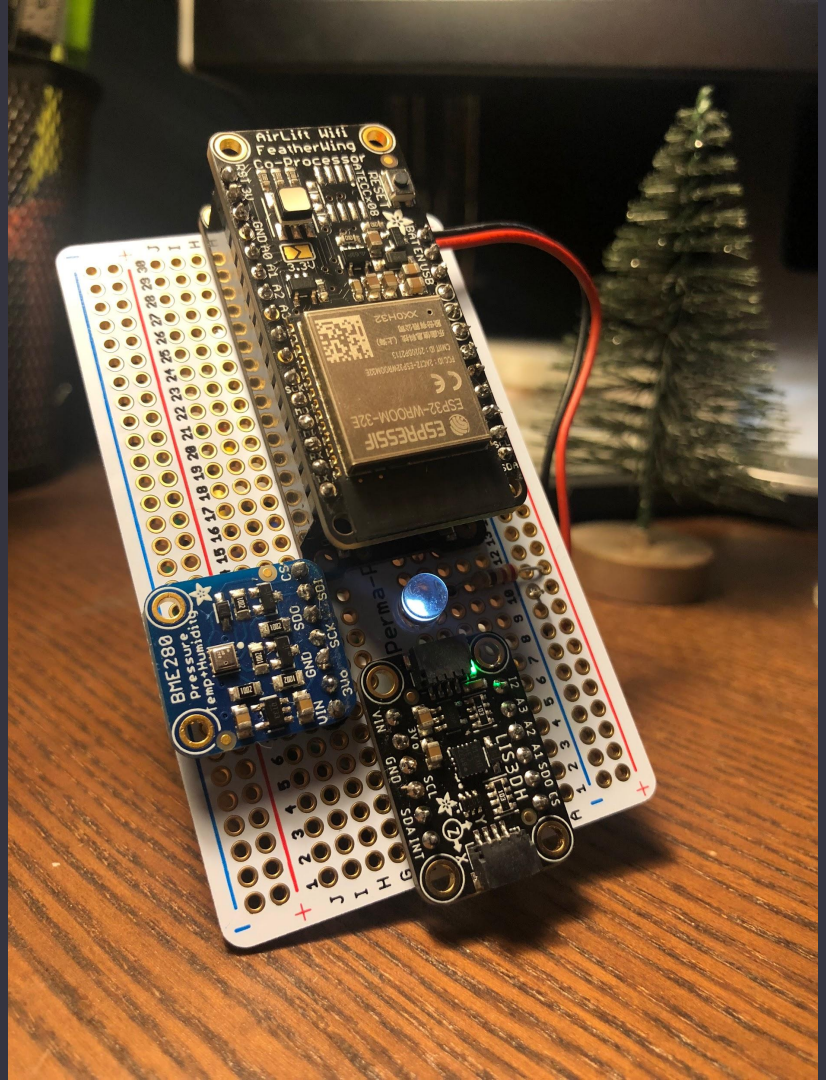
ALVIN
60°
45°


AirLift Wifi
FeatherWing
Co-Processor
ESPRESSIF
ESP32-WROOM-32E
BME280
Pressure
Temp+Humidity
LIS3DH